Author Contributions Checklist Form

This form documents the artifacts associated with the article (i.e., the data and code supporting the computational findings) and describes how to reproduce the findings.

Part 1: Data

☐ This paper **does not** involve analysis of external data (i.e., no data are used, or the only data are generated by the authors via simulation in their code).

☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Abstract

1) The file Arrythmia_preprocessed.csv contains the dataset we analyzed in section 4.2. It is about distinguishing between the presence and absence of cardiac arrhythmia. The original data, freely available online at https://archive.ics.uci.edu/ml/datasets/arrhythmia has been preprocessed to enable sampling from PyStan.
2) The file Voice_preprocess.csv contains the dataset we analyzed in section 4.2. It is about assess whether voice rehabilitation treatment lead to phonations considered 'acceptable' or 'unacceptable'(binary classification problem). The original data, freely available online at https://archive.ics.uci.edu/ml/datasets/LSVT+Voice+Rehabilitation has been preprocessed to enable sampling from PyStan.
3) The file Parkinson_preprocess.csv contains the dataset we analyzed in section 4.2. It is about assessing the Parkinson disease(classification). The original data, freely available online https://archive.ics.uci.edu/dataset/470/parkinson+s+disease+classification has been preprocessed to enable sampling from PyStan.
4) The file laekaemia.csv contains the dataset analyzed in section S.2.4 of the Supplement. It is about the survival of Leukaemia patients.

Availability

☒ Data **are** publicly available.

☐ Data **cannot be made** publicly available

If the data are publicly available, see the *Publicly available data* section. Otherwise, see the *Non-publicly available dat*a section, below.

Publicly available data

☒ Data are available online at: https://anonymous.4open.science/r/Mixture_IS-A64E/Data

☒ Data are available as part of the paper's supplementary material.

☐ Data are publicly available by request, following the process described here:

☐ Data are or will be made available through some other mechanism, described here:

Non-publicly available data

Discussion of lack of publicly available data:

Description

File format(s)

☒ CSV or other plain text:

☐ Software-specific binary format (.Rda, Python pickle, etc.):

☐ Standardized binary format (e.g., netCDF, HDF5, etc.):

☐ Other (described here):

Data dictionary

☐ Provided by the authors in the following file(s):

☐ Data file(s) is (are) self-describiing (e.g., netCDF files)

☐ Available at the following URL:

1) Arrythmia_preprocessed.csv

- y: presence of any form cardiac arrhythmia. Hence it has been dichotomized with respect to the original dataset.
- X0 to X207: attributes of patients

2) Voice_preprocessed.csv

- y: voice rehabilitation treatment outcome (acceptable or unacceptable)
- X0-X311: attributes of measurements

3) Parkinson_preprocessed.csv

- y: presence of Parkinson
- X0-X754: attributes of measurements

4) Leukaemia.csv

- y: survival of patient with Leukaemia after 50 weeks
- ones: intercept
- wbc: count of white blood cells of patients
- ag: outcome of a test related to white blood cell characteristics.

Additional information (optional)

All the datasets are available inside the Data folder of the repository. The stack-loss dataset can be downloaded directly from the statsmodels.api library of Python

Part 2: Code

Abstract

The repository has codes in three different programming languages: R, Python and Julia. The codes inside the Python_codes folder allow one to generate some of the estimators present in the paper for the conjugate Gaussian model under synthetic data generation, for the Stack loss dataset and for the for the Leukemia dataset for the Logistic model. The scripts inside the R_codes folder allow one to generate the estimators for the high-dimensional logistic regression models. The scripts present in the Julia folder are intended to be used for some of the most computation-heavy experiments of the paper.

Description

Code format(s)

☒ Script files
- ☒ R ☒ Python ☐ Matlab
- ☒ Other: Julia

☐ Package
- ☐ R ☐ Python ☐ MATLAB toolbox
- ☐ Other:

☒ Reproducible report
- ☐ R Markdown ☒ Jupyter notebook
- ☐ Other:

☐ Shell script

☐ Other (described here):

Supporting software requirements

Version of primary software used

1) When using PyStan, use PyStan 2.19.1.1 coupled with Python 3.6
2) If not using PyStan(conjugate models), Python 3.10
3) Julia 1.8.1
4) R version 4.2.3

5) Rstudio 2023.03.1+446

Libraries and dependencies used by the code.

- Python libraries:

  Numpy 1.19.5
  Pandas 1.1.5
  Scipy 1.5.4
  Pystan 2.19.1.1
  Matplotlib 3.3.4

- Julia libraries:

  Distributions v0.25.87

- R libraries:

  Rstan 2.21.8

Supporting system/hardware requirements (optional)

Parallelization used

☐ No parallel code used
☒ Multi-core parallelization on a single machine/node
Number of cores used: 50
☐ Multi-machine/multi-node parallelization
Number of nodes and cores used:

License

☒ MIT License (default)
☐ BSD
☐ GPL v3.0
☐ Creative Commons
☐ Other (described here):

Additional information (optional)

The parallelization was done only for generating the benchmark values for the high-dimensional logistic regression examples of section 4.2 and for the experiments of section 4.1.1. For all the other codes a standard PC is more than sufficient to run the experiments.

Part 3: Reproducibility workflow

Scope

The provided workflow reproduces:

☐ Any numbers provided in text in the paper

☐ The computational method(s) presented in the paper (i.e., code is provided that implements the method(s))

☐ All tables and figures in the paper

☒ Selected tables and figures in the paper, as explained and justified here:

Given the codes provided, it is trivial to generate the missing table/plots. Since there are quite a few in the paper, we selected a subset of them for conciseness.

Workflow details

Location

The workflow is available:

☒ As part of the paper's supplementary material

☒ In this Git repository: https://anonymous.4open.science/r/Mixture_IS-A64E

☐ Other:

Format(s)

☐ Single master code file

☐ Wrapper (shell) script(s)

☒ Self-contained R Markdown file, Jupyter notebook, or other literate programming approach

☒ Text file (e.g., a readme-style file) that documents workflow

☐ Makefile

☐ Other (more detail in 'Instructions' below)

Instructions

Find instructions for usage in README.md file.

Expected run-time

Approximate time needed to reproduce the analyses on a standard desktop machine:

☐ <1 minute

☒ 1-10 minutes

☒ 10-60 minutes

☐ 1-8 hours

☐ >8 hours

☒ Not feasible to run on a desktop machine, as described here:

Most of the codes can be run on a standard machine, while the generation of benchmark values for the high-dimensional logistic models of section 4.2 and the experiment of section 4.1.1 necessitates a machine with at least 24 physical cores to be performed in a reasonable timeframe.

Additional documentation (optional)

Notes (optional)